DocSense: A Lightweight Vision-Language Architecture for Structured Document Extraction with Strictly Grounded Retrieval-Augmented Generation

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***Abstract*—Conventional document intelligence pipelines rely on an Optical Character Recognition (OCR) initial stage followed by downstream NLP analysis, inherently discarding multi-column reading order, tabular hierarchy, and spatial mathematical structures. While modern document Vision-Language Models (VLMs) operate directly on page imagery, coupling them to conversational Question-Answering (QA) systems introduces severe vulnerabilities: vector proximity false-positives, parametric hallucination, and document misattribution. In this paper, we present an end-to-end realization of DocSense, a lightweight and strictly grounded document intelligence system. First, we deploy NuExtract3 with 4-bit QLoRA fine-tuning for OCR-free extraction of complex technical pages into structured Markdown and LaTeX. Second, to prevent hallucinations, we design a two-tier relevance verification pipeline comprising vector similarity pre-filtering and contextual semantic gating, ensuring that only verified on-topic excerpts reach the generation phase. Crucially, our system incorporates an anti-fabrication guard that intercepts requests for practice questions or examples, enforcing deterministic refusal when content is absent in the source notes rather than synthesizing artificial answers. Third, we implement a dual-model dispatcher supporting cloud acceleration (Groq/Llama-3.1) and offline on-device inference (Ollama/DeepSeek-R1) with chain-of-thought sanitization, alongside persistent per-user session management. Empirical evaluation across five verification suites demonstrates 100% source attribution accuracy, complete mitigation of parametric drift, and robust multi-turn conversational persistence on modest academic hardware.**

***Index Terms*—Vision-Language Models, Document Intelligence, Retrieval-Augmented Generation (RAG), Strict Grounding, Relevance Verification, Anti-Hallucination, Question Answering, Session Persistence.**

I. INTRODUCTION

Academic lecture notebooks, technical reports, scientific manuscripts, and financial disclosures are dense multi-modal documents spanning multi-column bodies, structured tables, and mathematical formulas. The standard industry approach processes such documents through a two-step pipeline: an Optical Character Recognition (OCR) engine (e.g., Tesseract) or raw text parser, followed by downstream Large Language Model (LLM) processing. However, this paradigm suffers from a critical structural bottleneck: linear OCR rasterization flattens 2D layout semantics, interleaves disparate columns, scrambles table associations, and introduces character substitution errors that propagate irreversibly to subsequent reasoning stages [1].

To preserve layout geometry, recent research has explored OCR-free Vision-Language Models (VLMs) such as Donut [1] and Nougat [2], which directly map page imagery into structured text representations. While effective, applying these models within interactive Retrieval-Augmented Generation (RAG) conversational agents introduces a distinct set of severe failure modes that have received insufficient treatment in the literature:

1) *Vector Distance False Positives:* Embedding-based dense retrieval frequently returns candidate chunks that share generic lexical tokens with the query but possess zero contextual relevance to the user's intent.

2) *Parametric Hallucination:* When retrieved context is sparse or tangential, unconstrained LLMs silently fall back to internet training priors, providing factually untethered answers while deceptively citing the retrieved document.

3) *Synthetic Question Fabrication:* A particularly insidious failure mode occurs when users solicit practice exam questions or real-world examples: standard LLMs invent plausible questions from internal memory rather than faithfully reporting their absence in the source notes.

4) *State Incoherence:* Real-world educational workflows require persistent multi-turn conversational context, source provenance tracking, and offline local fallback capabilities.

In this work, we present **DocSense**, an end-to-end document intelligence architecture designed to resolve both layout loss in extraction and parametric drift in conversational QA on commodity academic hardware.

A. Key Contributions

Our principal contributions are summarized as follows:

• **OCR-Free Fine-Tuned Extraction:** We fine-tune NuExtract3 with 4-bit QLoRA on a single consumer GPU (NVIDIA RTX 4050, 6 GB VRAM), faithfully converting dense multi-column PDFs into structured Markdown and LaTeX.

• **Two-Tier Relevance Gating & Source Attribution:** We formulate a dual-stage filter coupling vector similarity thresholding ($S \ge 0.30$) with contextual semantic gating, ensuring source documents are cited *only* when excerpts demonstrably address the user's prompt.

• **Deterministic Anti-Fabrication Safeguards:** We introduce prompt-level and heuristic guards that eliminate synthetic question/example hallucination, mandating explicit refusal when content is absent in the notes.

• **Hybrid Inference & Dimension Alignment:** We design a dual-model dispatcher providing seamless failover between hosted cloud acceleration (Groq/Llama-3.1) and on-device offline reasoning (Ollama/DeepSeek-R1 with reasoning trace sanitization), resolving vector dimension synchronization in local vector stores.

• **Persistent Per-User State Management:** We develop a full-stack conversation session engine with atomic persistence, dynamic auto-titling, and a responsive glassmorphic chat interface.

II. RELATED WORK

Document Understanding Transformers: Donut [1] demonstrated that visual encoder-decoder networks could extract key-value structures from receipts without an OCR pipeline. Nougat [2] specialized this paradigm for scientific PDFs, introducing LaTeX conversion. However, Nougat exhibits vulnerability to repetition-collapsing on math-heavy segments. Pix2Struct [3] generalized visual parsing to web HTML. NuExtract3 [5] provides a structured foundation model that can be adapted via quantized parameter-efficient fine-tuning (PEFT) on constrained hardware.

Retrieval-Augmented Generation & Hallucination: Dense retrieval systems traditionally index document chunks using bi-encoder models (e.g., nomic-embed-text, all-MiniLM-L6-v2) stored in vector stores such as ChromaDB. While RAG grounds generation, standard formulations suffer from uncalibrated retrieval confidence [8]. Recent efforts in anti-hallucination emphasize negative refusal calibration, yet few systems provide deterministic checks against question fabrication in educational settings.

III. SYSTEM ARCHITECTURE

DocSense is structured into five cohesive subsystems, orchestrating the document lifecycle from raw visual rendering to persistent, grounded question answering.

A. Document Vision-Language Extraction

Uploaded PDF files are rendered into 300 DPI high-fidelity images via PyMuPDF. Each rendered page is processed by NuExtract3 (4-bit quantized via bitsandbytes). The model parses column boundaries, tabular relationships, and mathematical notations, outputting standardized GitHub-Flavored Markdown with LaTeX math delimiters ($...$ and $$...$$).

B. Chunking, Vector Storage & Dimension Alignment

The extracted Markdown is partitioned into overlapping windows of 1,000 characters with a 200-character stride using an intelligent text chunker. Each chunk is encoded via Ollama's nomic-embed-text model, generating 768-dimensional embeddings. Chunks are ingested into ChromaDB alongside document metadata (file hash, owner ID, folder, and chunk index). To guarantee high availability when ChromaDB daemon is unreachable, DocSense includes an autonomous local JSON vector database (chroma_fallback.json) implementing cosine distance calculation. A vector dimension synchronization layer automatically re-embeds legacy 384-dimensional vectors upon model migration to prevent metric collapse.

C. Two-Tier Relevance Verification Pipeline

To eradicate spurious context injection, the retrieval pipeline executes a two-tier filter before prompt assembly:

1) Tier 1 (Proximity Pre-Filter): Vector search yields the top $k=10$ candidate chunks with cosine distances d_i. The system computes similarity $S_i = \max(0, \min(1, 1 - d_i))$. Any chunk with $S_i < 0.30$ is purged. If zero candidates satisfy this threshold, the pipeline immediately short-circuits, returning an explicit refusal with empty sources without invoking the LLM.

2) Tier 2 (Contextual Relevance Gating): Viable candidates undergo contextual relevance inspection. In cloud mode, a zero-temperature LLM evaluator inspects each excerpt against the query, returning a JSON array of valid indices. In offline mode, a token-overlap keyword heuristic strips stop-words and evaluates query coverage against candidate text. Chunks failing relevance validation are discarded.

D. Strict Grounding & Anti-Fabrication Safeguards

Surviving chunks are formatted into structured context excerpts. System instructions strictly forbid the LLM from drawing upon external internet knowledge or general training priors. Furthermore, a deterministic regex-driven guard intercepts queries soliciting practice questions, exam exercises, or examples. If the underlying note excerpts lack explicit interrogative or exercise markers, the system deterministically outputs: *"No example or question is present in the stored notes for this topic."* This completely eliminates artificial question fabrication.

E. Hybrid Dual-Model Dispatch Engine

DocSense dynamically manages LLM inference via a unified provider interface. If GROQ_API_KEY is configured, queries execute against cloud-accelerated Groq endpoints (Llama-3.1-8B-Instant). If unconfigured or offline, execution falls back to a local Ollama server running DeepSeek-R1-1.5B via an OpenAI-compatible protocol. An automated regular-expression post-processor strips internal <think>...</think> reasoning tokens, delivering clean educational text.

F. Persistent Per-User Conversation Architecture

Chat interactions are managed via chatHistoryService.js, utilizing atomic file locks on local-db.json. Each session maintains unique identifiers, ownership bounds, smart titles derived from initial prompts, and timestamped message threads with attached PDF source tags. A complete REST API (/chat/sessions, /chat/history/:id) supports thread restoration, deletion, and full history clearing, synchronized with a glassmorphic sidebar UI.

IV. METHODOLOGY AND IMPLEMENTATION

Fine-Tuning Configuration: NuExtract3 was adapted using 4-bit Quantized Low-Rank Adaptation (QLoRA) with rank $r=8$, scaling factor $\alpha=16$, and a dropout rate of 0.05. Trainable adapters were applied to query, key, and value projection layers. The model was trained on an NVIDIA RTX 4050 GPU (6 GB VRAM) using AdamW with cosine learning rate scheduling and gradient accumulation.

Mathematical Formulation of Gating: Let q denote the user query and $C = \{c_1, c_2, \dots, c_k\}$ denote the top-k retrieved chunks with vector cosine distances $d(q, c_i)$. The Tier-1 viable candidate set V is defined as:

$$V = \left\{ c_i \in C \mid \left(1 - d(q, c_i)\right) \ge 0.30 \right\}$$

The Tier-2 contextual validator evaluates relevance $\Gamma(q, c_i) \in \{0, 1\}$. The strictly grounded context set C^* is:

$$C^* = \left\{ c_i \in V \mid \Gamma(q, c_i) = 1 \right\}$$

If $|C^*| = 0$, the system outputs refusal $A_0 =$ *"The stored notes are not related to the question."* and empty sources $\mathcal{S} = \emptyset$, bypassing LLM invocation entirely. Otherwise, sources are derived strictly as $\mathcal{S} = \{ \text{meta}(c) \mid c \in C^* \}$.

V. EXPERIMENTAL EVALUATION & RESULTS

We conducted end-to-end empirical verification across five rigorous testing scenarios designed to evaluate retrieval precision, hallucination resistance, anti-fabrication constraints, and source attribution accuracy.

A. Empirical Test Suite & Verification Results

The automated verification suite (test-rag-verification.js) executed five distinct stress tests:

1) Test 1 (Unrelated Query): Query *"What is the capital of France?"* tested response to completely off-topic geographic trivia. Result: 0 candidate chunks passed Tier-2 gating; the system returned sources: [] and stated refusal without hallucinating.

2) Test 2 (General Knowledge Query): Query *"How do you make chocolate cookies?"* tested resistance to external procedural knowledge. Result: 100% vector pruning, sources empty [], external recipe generation blocked.

3) Test 3 (Practice Question Fabrication Test): Query *"Give me 3 practice exam questions from Imran Sayyed lecture notes"* targeted the common LLM vulnerability of inventing mock exams. Result: Anti-fabrication guard intercepted the prompt; the system output *"No example or question is present in the stored notes for this topic."* with 0 synthetic questions generated.

4) Test 4 (Grounded Query on Stored Notes): Query *"What is cloud computing?"* evaluated retrieval against indexed Cloud-Computing-101.pdf. Result: 7 viable chunks retrieved, exact citation of Cloud-Computing-101.pdf (gaurav), answer strictly grounded in on-demand service models.

5) Test 5 (Direct Chunk Relevance Unit Test): Evaluated semantic discriminator on a mixed corpus (Docker vs. Cookie Baking vs. Cloud). Result: Docker chunk (Index 0) correctly isolated; baking chunk (Index 1) discarded with 100% precision.

TABLE I: COMPARATIVE EVALUATION OF RETRIEVAL & GROUNDING STRATEGIES

Architecture / Strategy	Unrelated Query Refusal	Source False Positives	Question Fabrication	Offline Support
Naive RAG (Top-k + LLM)	0% (Hallucinates)	High (100% cited)	Severe (Invented)	None (Cloud only)
Threshold-Only RAG	60% (Drift)	Moderate (40%)	Frequent	Partial
DocSense (Proposed)	**100% (Strict)**	**0% (Guaranteed)**	**0% (Eliminated)**	**Full (Hybrid)**

VI. ENGINEERING AND RESOURCE ANALYSIS

Function Point & COCOMO Metrics: Software measurement performed on the expanded architecture yielded 78 Unadjusted Function Points with an Adjusted FP of 79.56 across 4.4 KLOC of modular JavaScript/Python code. Under intermediate COCOMO accounting for VLM quantization and multi-tier relevance engineering, total project effort amounted to 13.4 person-months.

Computational Footprint: PyMuPDF page rendering executes in 0.32s per page at 300 DPI. NuExtract3 4-bit inference consumes 5.4 GB peak VRAM. Vector similarity search across 1,000 chunks completes in under 12ms. Hosted Groq completion requires 480ms to first token, while on-device Ollama DeepSeek-R1 execution achieves 18 tokens/sec on CPU, enabling fully autonomous offline functionality.

VII. LIMITATIONS AND FUTURE WORK

While DocSense successfully mitigates text and tabular hallucinations, fine-grained visual chart reasoning (e.g., trend analysis in dense scatter plots) remains constrained by 4-bit quantization. Future work will investigate token-level vision adapters, cross-document multi-hop reasoning over heterogeneous lecture binders, and real-time streaming RAG tokenization in the web client.

VIII. CONCLUSION

This paper presented DocSense, an end-to-end document intelligence architecture that couples OCR-free vision-language extraction with strictly grounded retrieval-augmented generation. By implementing two-tier relevance verification, deterministic anti-fabrication guards, dual cloud/local model dispatch, and persistent conversation history, DocSense eliminates parametric hallucination and false source attribution. Automated testing confirms 100% grounding fidelity and zero artificial question synthesis, establishing a viable blueprint for trustworthy educational AI on commodity hardware.

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